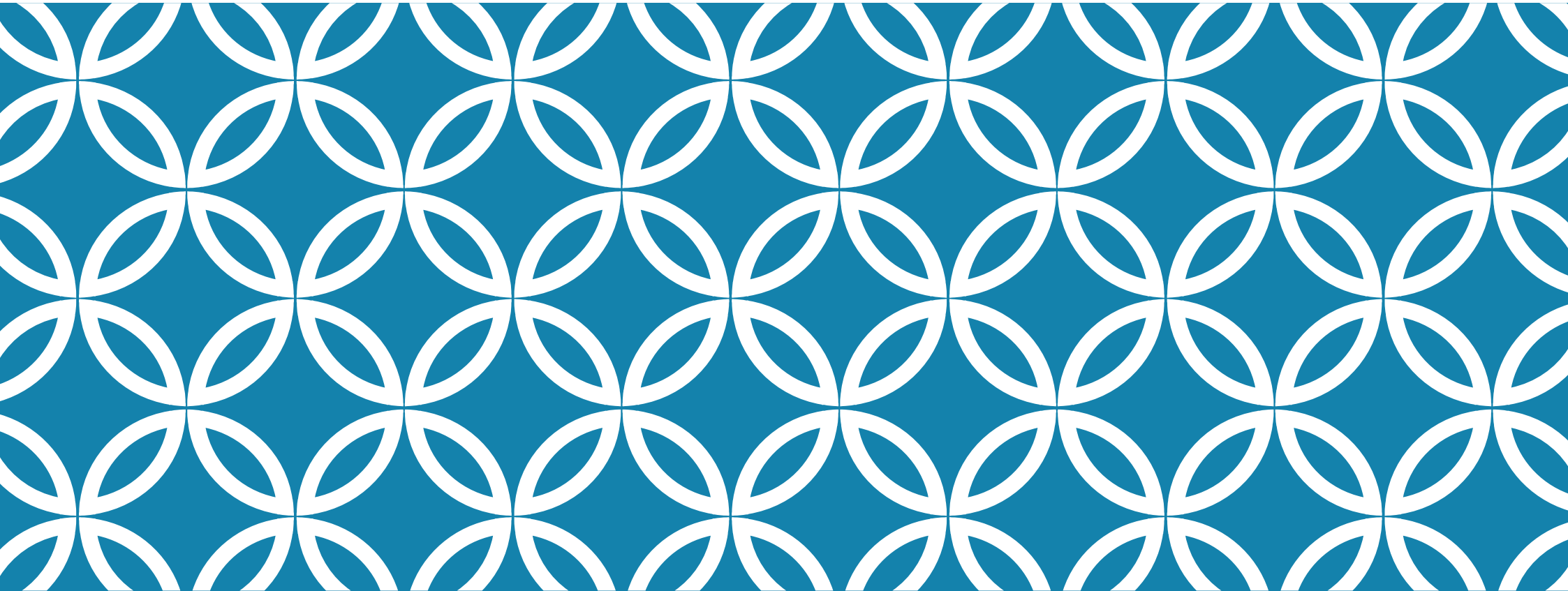
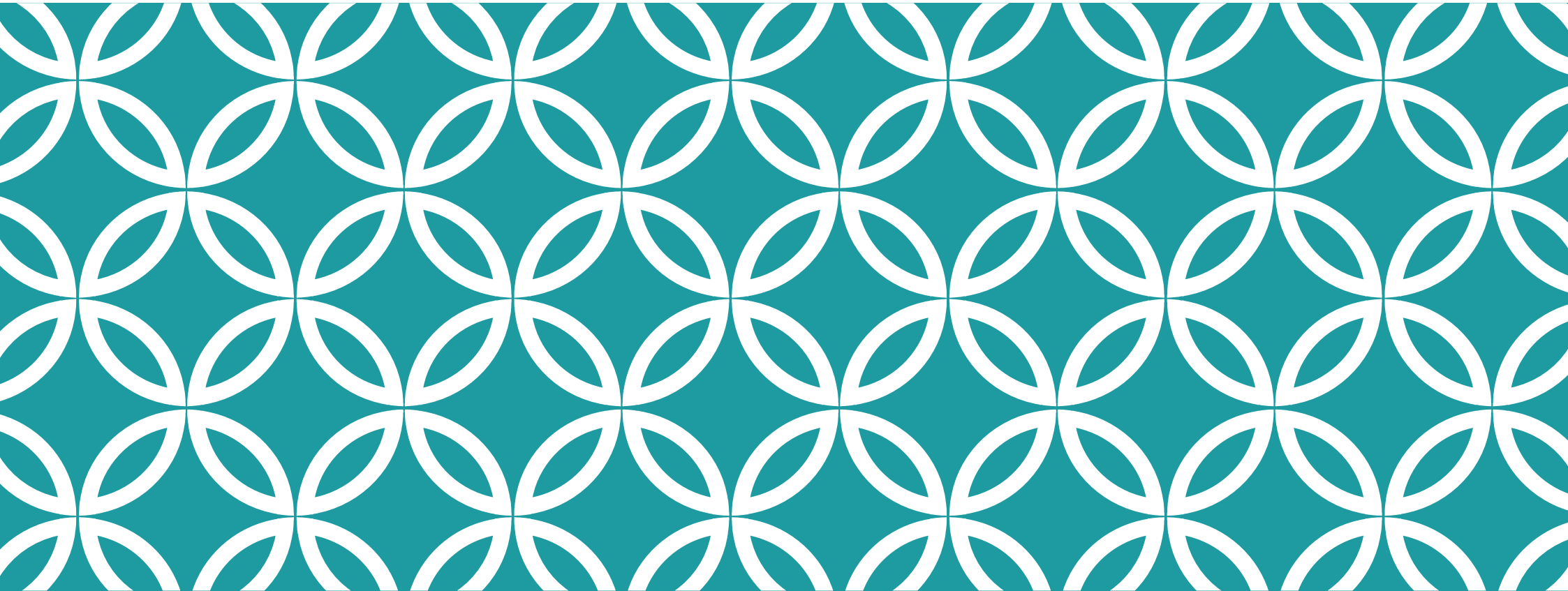




LAB 2

UNIVERSITY MANAGEMENT SYSTEM

Learning by doing



SQL SERVER MANAGEMENT STUDIO

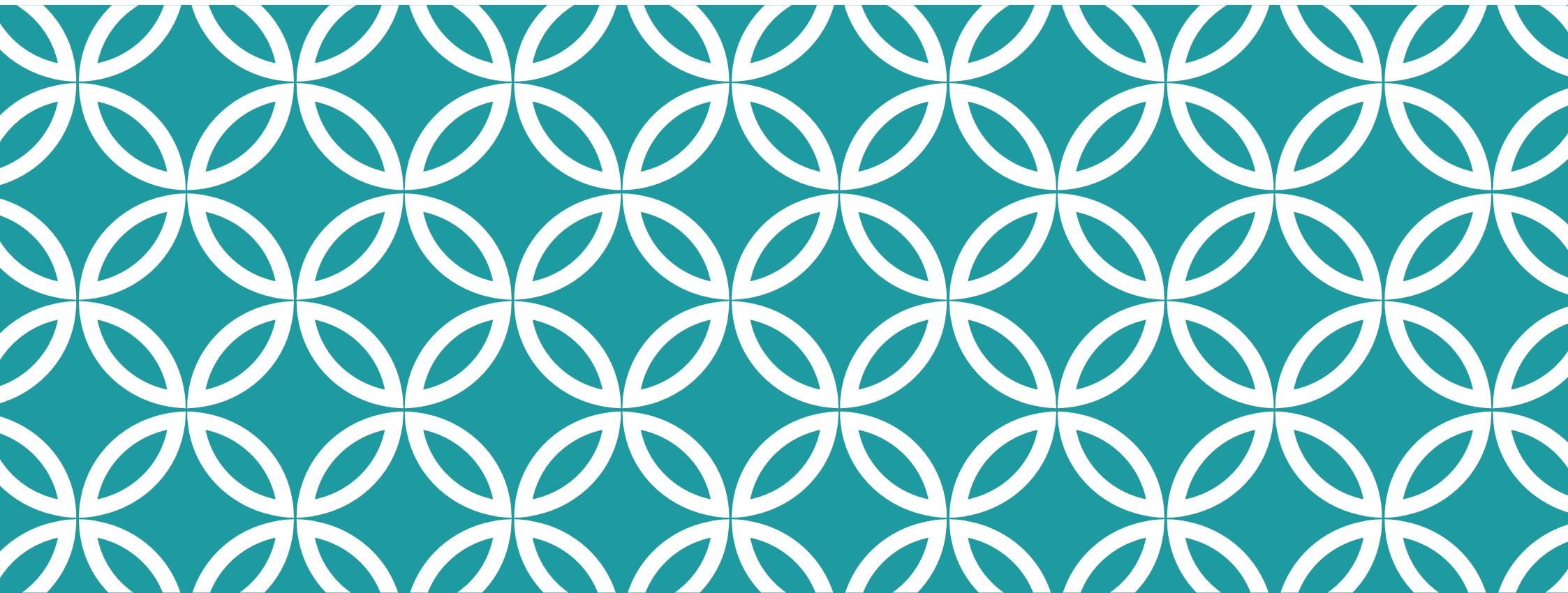
PART 1:

Managing database

CREATE ACCOUNT TABLE

1. Use database: **University**
2. Create table: **user**

	Column Name	Data Type	Allow Nulls
🔑	ID	int	☐
	username	varchar(50)	☒
	password	varchar(50)	☒
	role	varchar(20)	☒

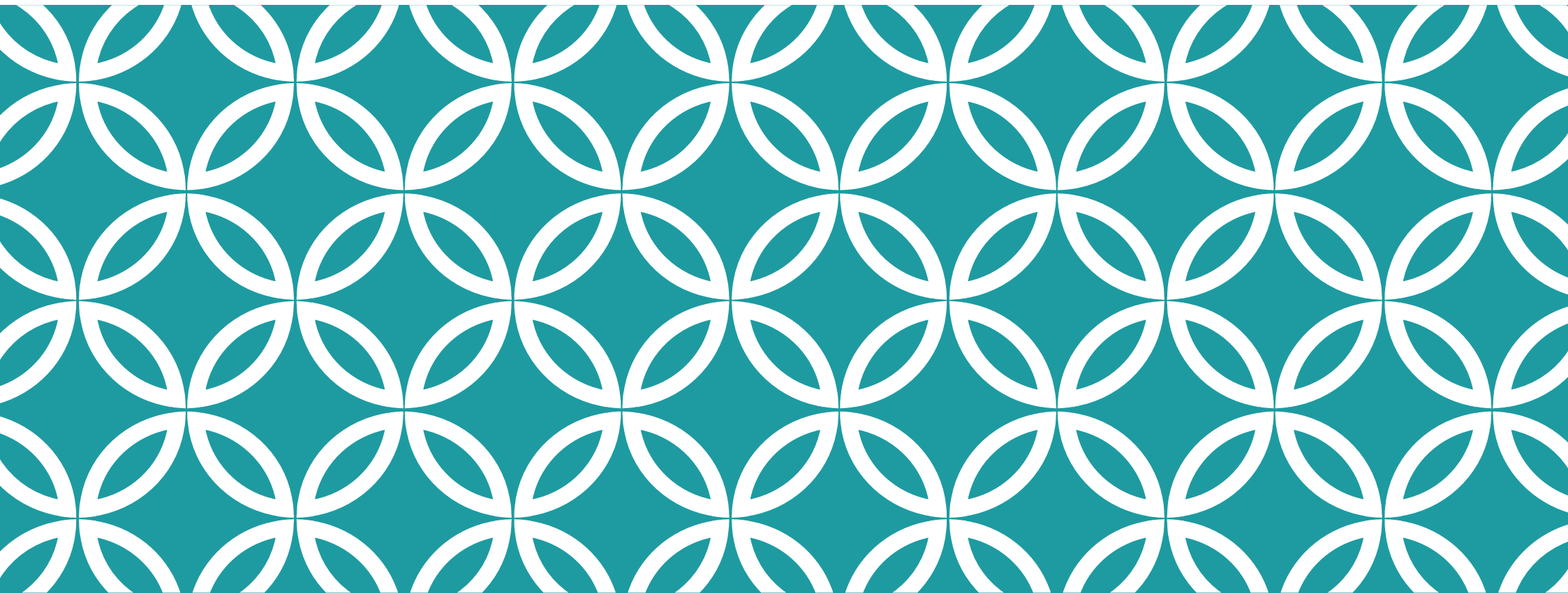


PART 2: VISUAL STUDIO

Application development

UNIVERSITY MANAGEMENT SYSTEM

1. New Project: **University Management System**
2. Add New Item: **ADO.NET Entity Data Model**
3. Build Solution to create data objects
4. Change the Form properties: **Text**

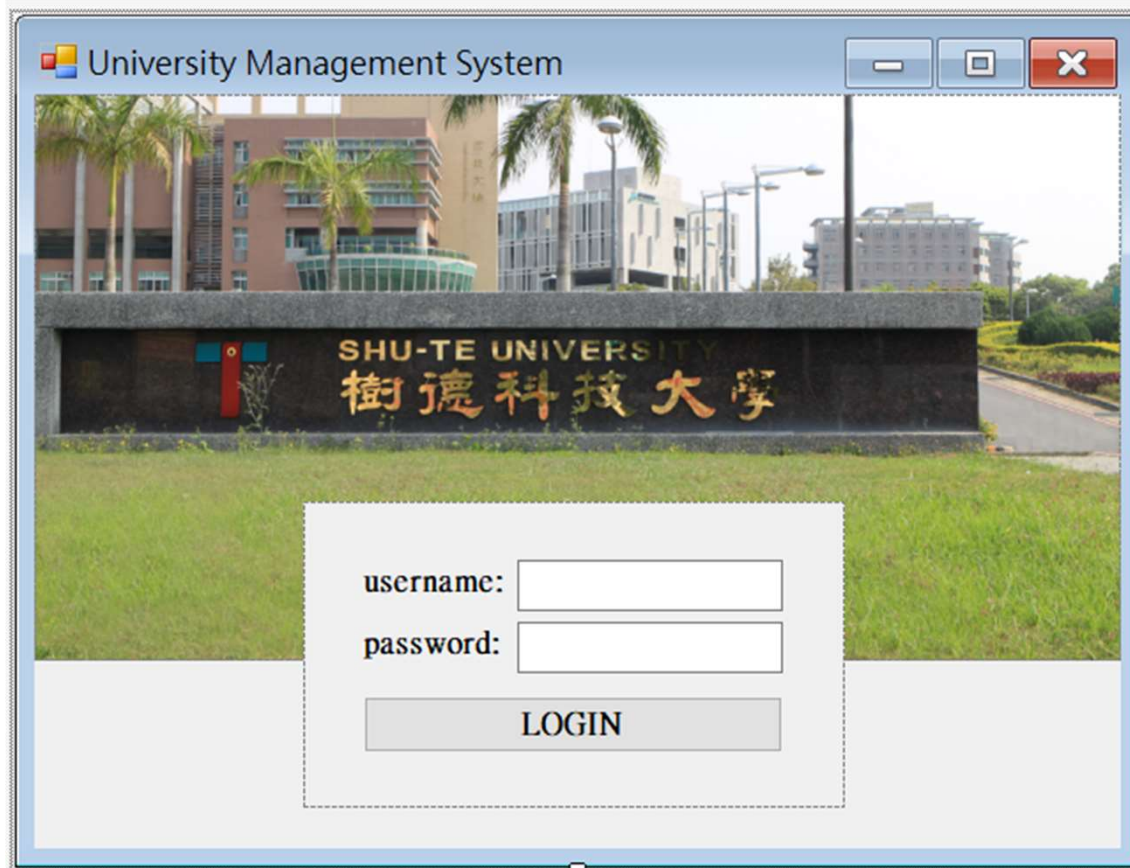


LOGIN FORM



LOGIN FORM

1. Form design by using Toolbox and Properties



The image shows a screenshot of a login form designed to look like a Windows application window. The window title bar reads "University Management System" and includes standard minimize, maximize, and close buttons. The main content area features a background image of a university campus with a large stone sign that reads "SHU-TE UNIVERSITY" and "樹德科技大學". Overlaid on this background is a light gray rectangular form with a dashed border. Inside this form, there are two labels: "username:" and "password:", each followed by a white text input field. Below these fields is a gray button labeled "LOGIN".

LOGIN FORM

1. Add an event handler for 'LOGIN' button click event

```
// login button
private void button1_Click(object sender, EventArgs e)
{
    // validate input
    if (string.IsNullOrEmpty(usernameTextBox.Text) ||
        string.IsNullOrEmpty(passwordTextBox.Text))
    {
        MessageBox.Show("username and password is required!", "Error", MessageBoxButtons.OK,
            MessageBoxIcon.Error);
        return;
    }

    // open database
    using (var db = new UniversityEntities())
    {
```


LOGIN FORM

1. Add an event handler for 'LOGIN' button click event

```
// check if user exists
var user = db.user.Where(u => u.ID == usernameTextBox.Text && u.password ==
    passwordTextBox.Text).FirstOrDefault();
if (user == null)
{
    MessageBox.Show("invalid username or password!", "Error", MessageBoxButtons.OK,
        MessageBoxIcon.Error);
    return;
}

// hide login form
usernameTextBox.Text = "";
passwordTextBox.Text = "";
this.Hide();
```

LOGIN FORM

1. Add an event handler for 'LOGIN' button click event

```
// open form based on role
switch (user.role)
{
    // administrator
    case "administrator":
        var adminForm = new AdminForm();
        adminForm.ShowDialog();
        break;

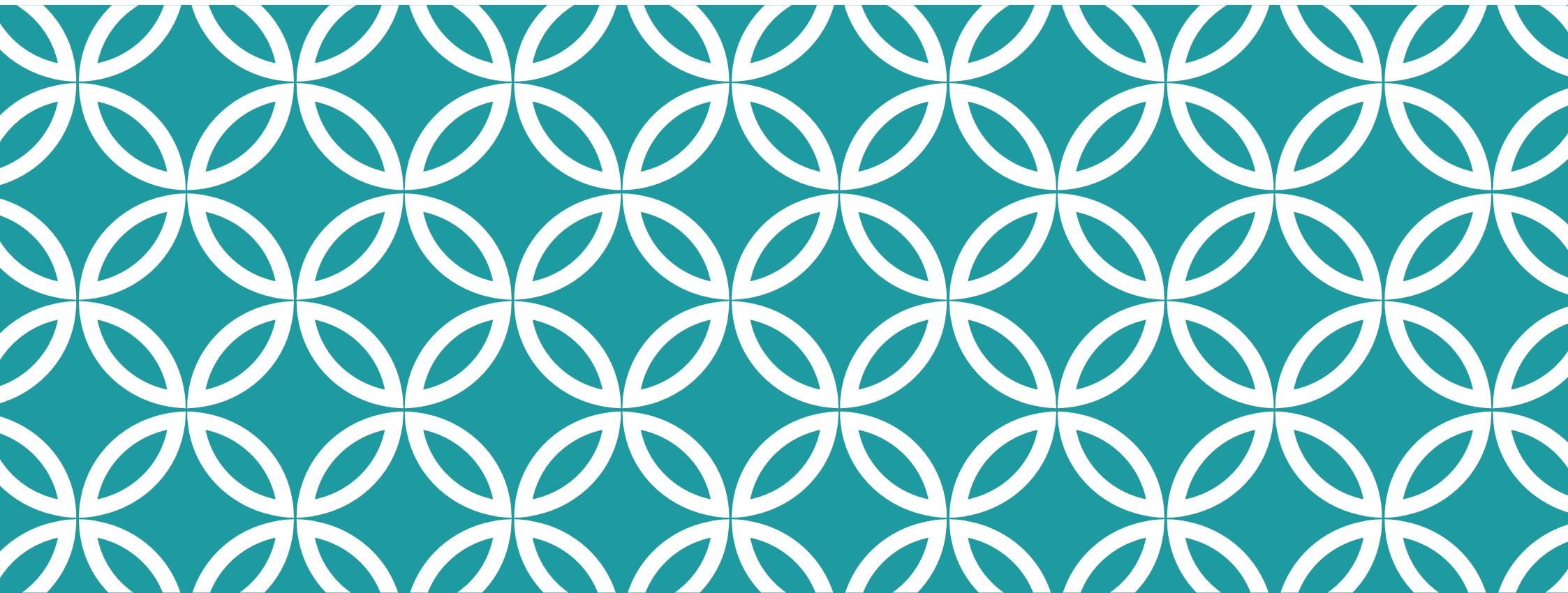
    // instructor
    case "instructor":
        var facultyForm = new FacultyForm();
        facultyForm.ShowDialog();
        break;
}
```

LOGIN FORM

1. Add an event handler for 'LOGIN' button click event

```
// student
case "student":
    var studentForm = new StudentForm();
    studentForm.ShowDialog();
    break;
}

// show login form
this.Show();
}
}
```



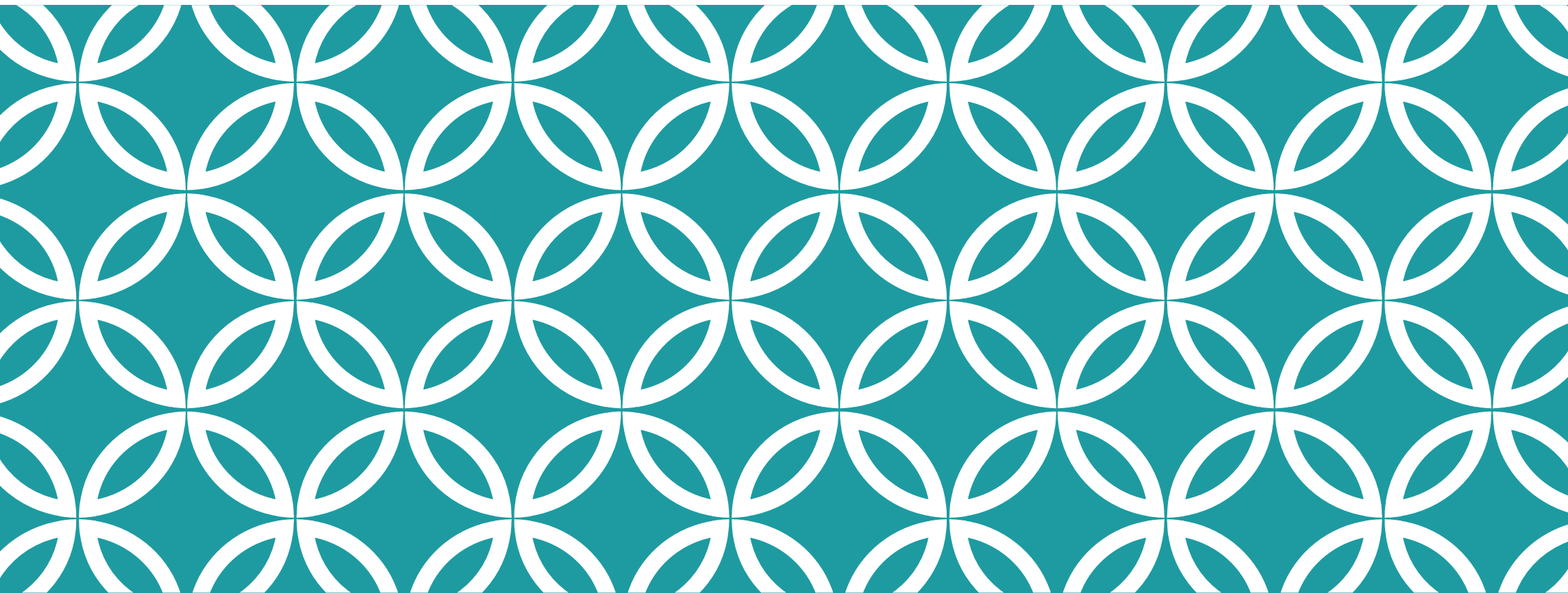
STUDENT FORM |

STUDENT FORM

1. Form design by using Toolbox and Properties

The screenshot shows a Windows application window titled "StudentForm". It features two tabs: "Profile" and "Enroll". The "Profile" tab is currently selected and contains four text boxes with labels: "dept name:", "ID:", "name:", and "tot cred:". The "Enroll" tab is visible but contains no content. To the right of the text boxes is a table with five columns: "course_id", "sec_id", "semester", "year", and "grade". The table body is currently empty and has a gray background. The window has standard Windows window controls (minimize, maximize, close) in the top right corner.

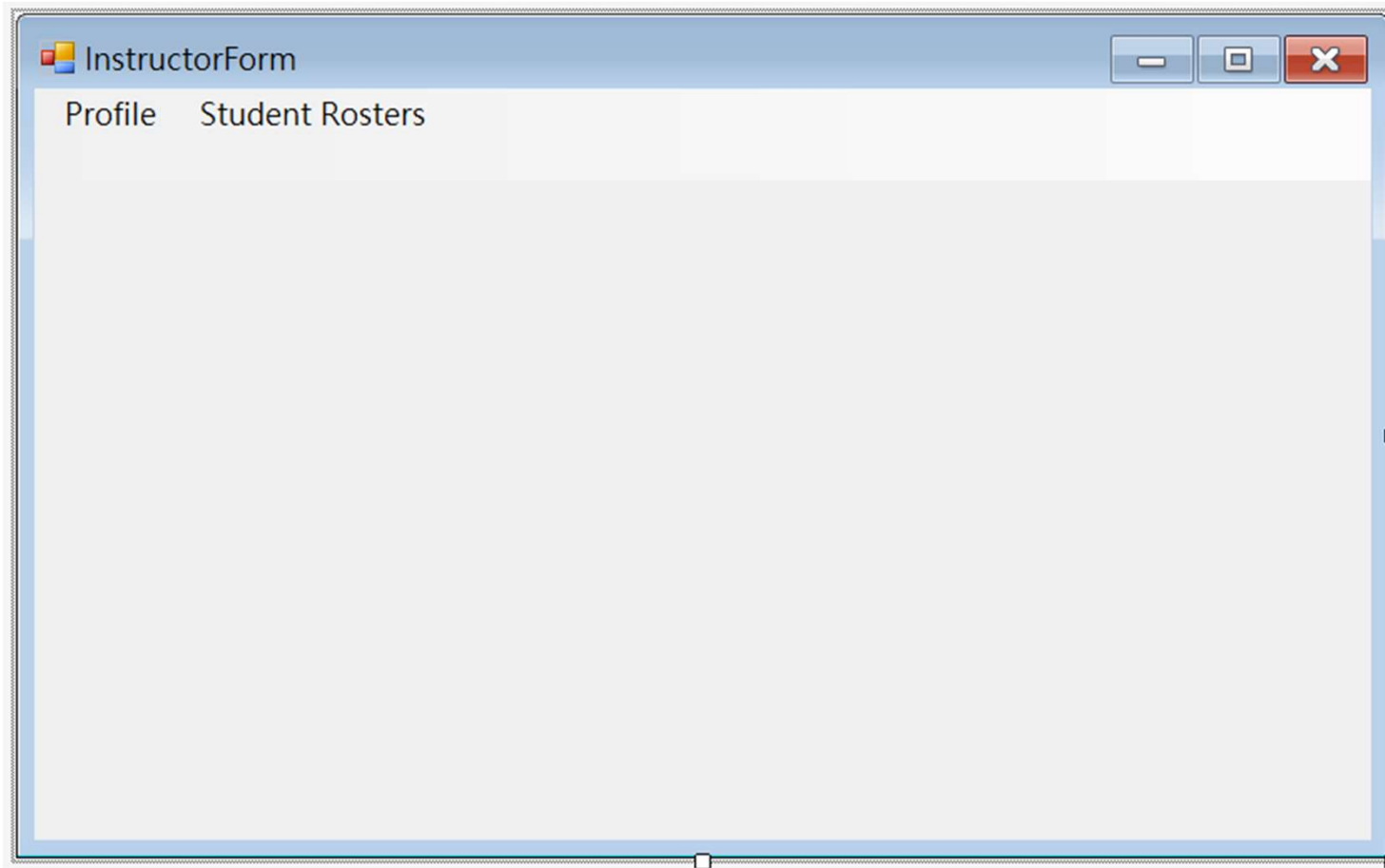
course_id	sec_id	semester	year	grade
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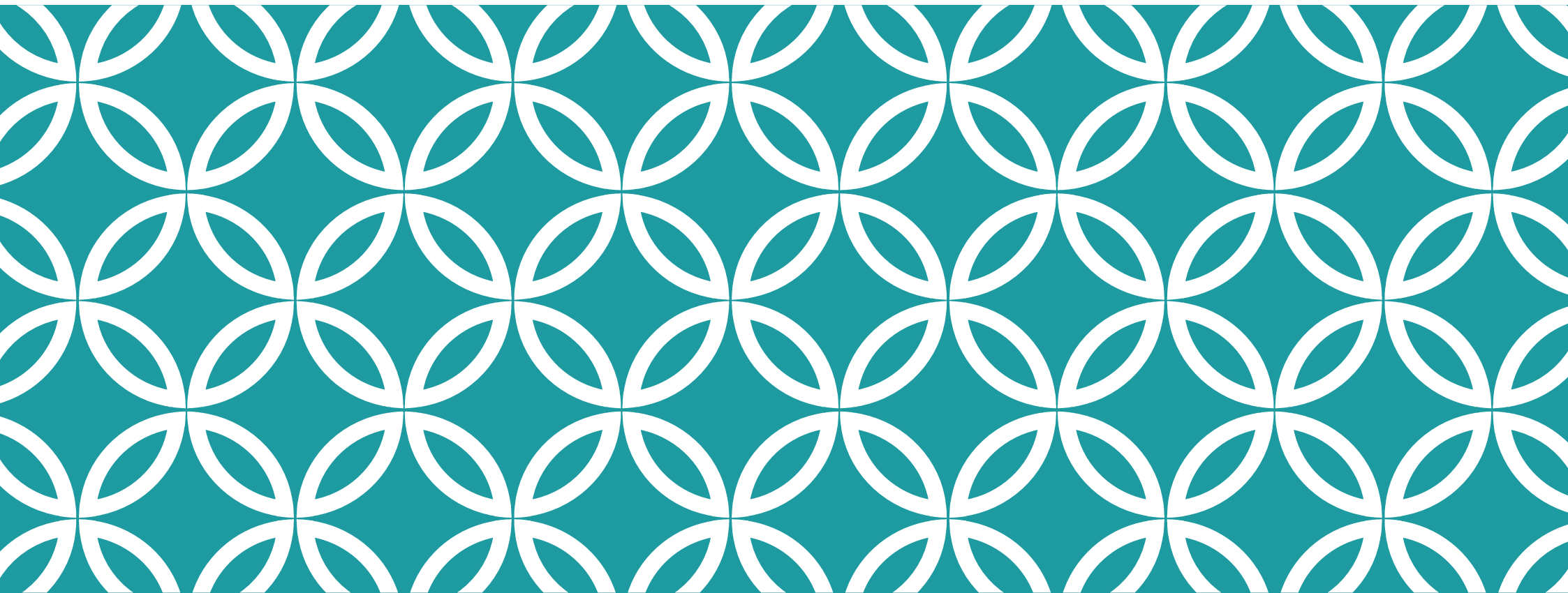


INSTRUCTOR FORM

INSTRUCTOR FORM

1. Form design by using Toolbox and Properties



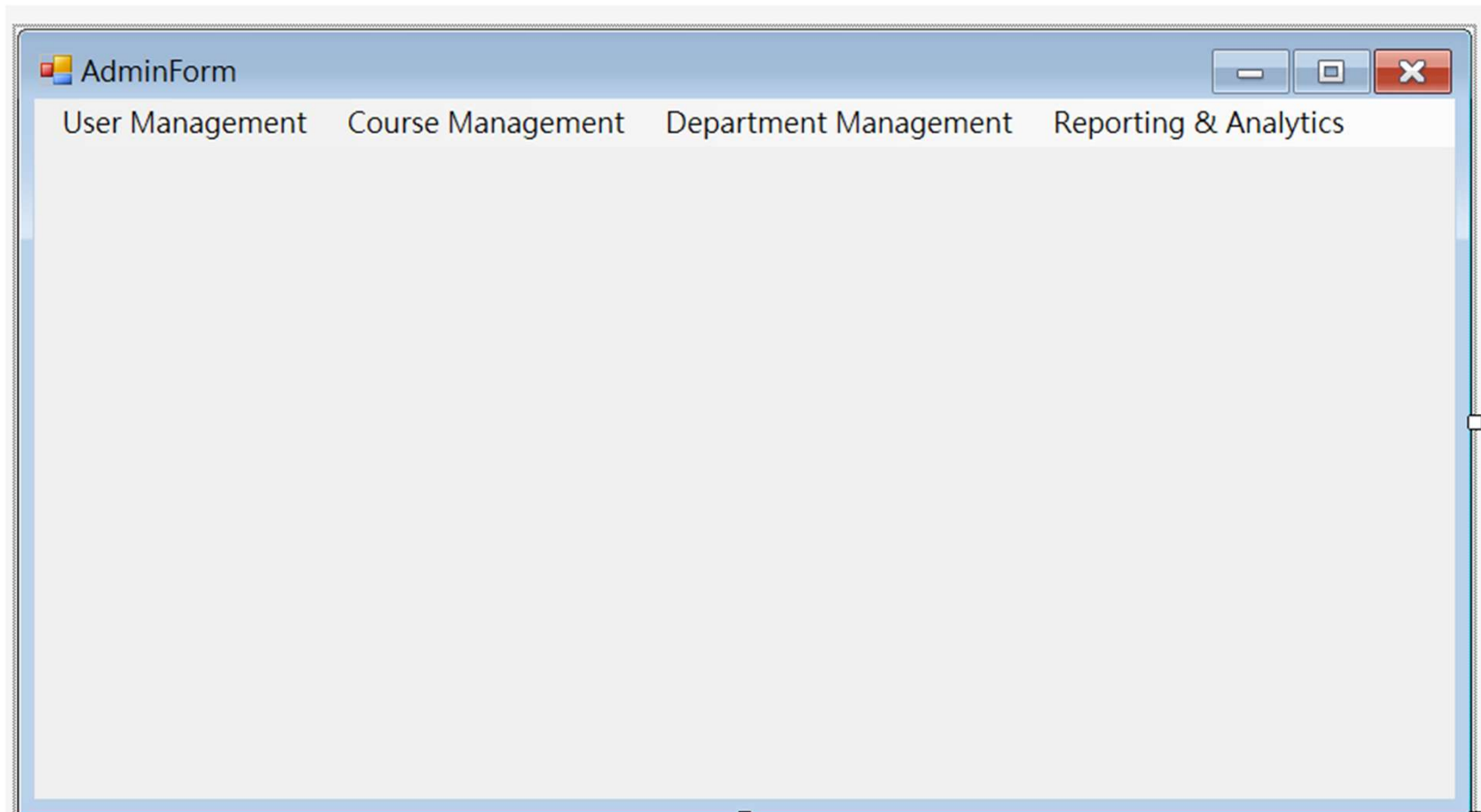


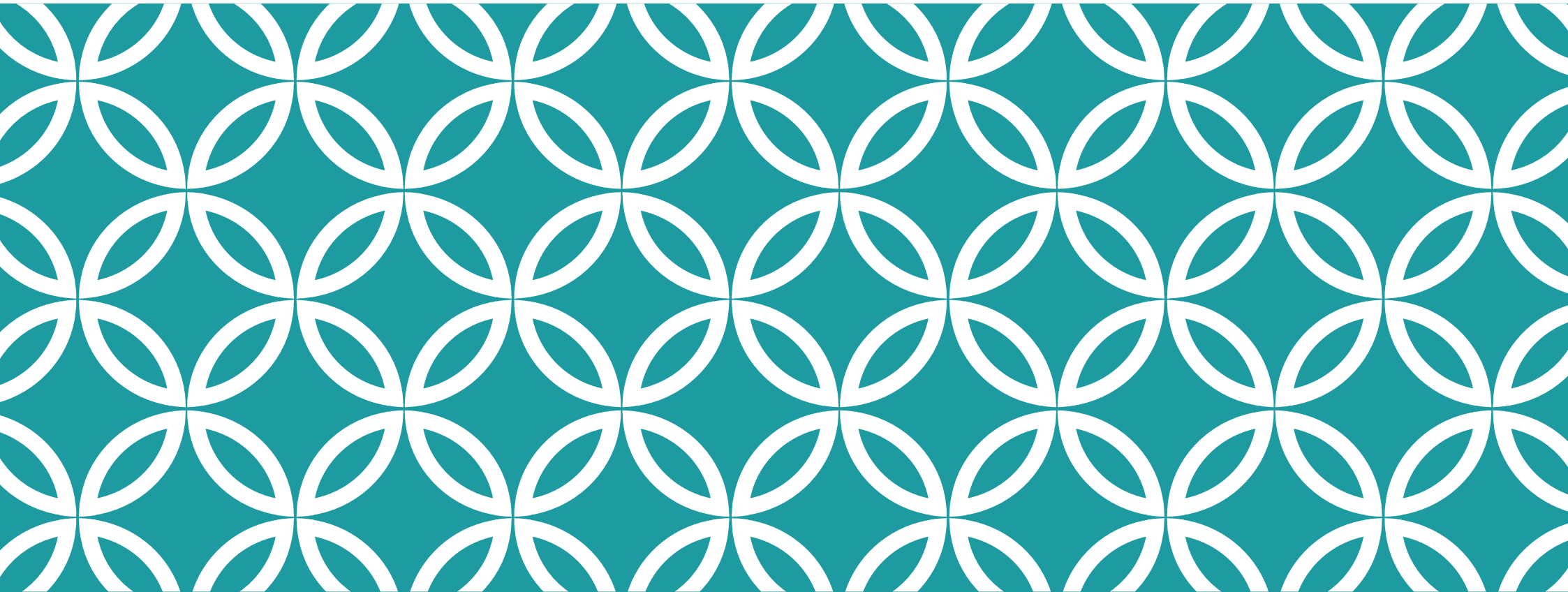
ADMIN FORM



ADMIN FORM

1. Form design by using Toolbox and Properties





Q & A

